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Task Description:	The focus of the BioS-ENDURES Consortium is innovation, acceleration, and implementation of space biology specific knowledge and technology centered on human-plant-microbiome relationships to enable a durable human presence in low Earth orbit and beyond. Studying fundamental relationships between humans, plants, and their microbiomes under spaceflight conditions will reveal new biological mechanisms that will address needs for sustainability and risk mitigation in space exploration. A more thorough understanding is needed of mechanisms underlying responses to space-relevant stressors (ionizing radiation, microgravity, circadian disruption, abiotics, and biotics) to both humans and plant food sources and what role microbiomes may play in shaping those responses. Relationships between microorganisms and humans in isolated and constrained environments such as space habitats must be better understood to enable successful long-term habitation in space. Human/animal, plant, and microbial biologists will work together to ensure an integrated view of the spaceflight biosphere by enhancing data acquisition, modeling, and testing. BioS-ENDURES has three thrust areas focusing on the effects of spaceflight stressors: 1. Develop monitoring capabilities to measure underlying molecular status (biomarkers) of humans, animal models, plants, and their associated microbial communities. Example projects: building radiation hardened and miniaturized sensor arrays for in situ real-time monitoring of biomarkers; developing cellular markers for different stressors in oxidative plant stress responses; creating an activity-based protein profiling capability for in-space assessment of microbial functional activity; developing multispectral image-based sensing of biological impacts. 2. Build models to predict human-plant-microbe robustness and interactions among organisms in space. Example projects: understanding relationships between gut microbiome functioning, host central-peripheral internal desynchrony, and blood-brain barrier permeability and brain functioning in circadian misalignment; modeling and validation of effects of spaceflight stressors on animal model/organoid function and resulting effects on microbiomes; modeling effects of enzyme kinetics in extreme temperature (high and low) on phytoplankton photosynthesis and mechanisms for adaptation; studying relationships between plant food sources and their resident microbiomes in the context of growth under spaceflight stressors. 3. Validate and apply understanding of human and plant health, including promoting beneficial human-plant-microbe interactions, to enhance health in space. Example projects: leveraging nasal microbial functioning to improve mental health; developing communities of soil microbes that interact with plants to convert CO2 to inorganic forms that can be precipitated and collected; identifying endophytic bacteria that promote crop yields and plant resilience to spaceflight stressors, including nutrient and water limitation. The BioS-ENDURES Consortium is built upon a collaboration between the University of Washington, Washington State University, the Pacific Northwest National Laboratory, and science and industry advisory boards. Consortium members will partner closely with NASA to align work with current and projected needs. The primary measure of success is the extent to which accomplishments assist NASA moving toward healthy human space exploration. Funding from NASA will support proof-of-principle demonstration projects each year to advance the science of the three thrusts, annual symposia tracks (some full consortium, some with tighter focus), and costs of physical testing. Impact assessment will be tied to seed funding being followed by longer-term external funding, published papers with new teams of collaborators, development of industry and community partnerships, translation of research to commercialization, and growth of the overall active consortium membership.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Humans, and the plants and microbes that we rely on, face significant challenges to mitigate stress and thrive self-sufficiently during long-term space habitation. BioS-ENDURES will support new research to broadly address the unique environment that space presents. • Development of biosensors for humans and plants that report on physiological status. • Testing with ionizing radiation sources and other abiotic stresses. • New examination of potential correlations and controls of human microbiomes on human performance and well-being, including biorhythm misalignment. • Use of microbes to enhance plant stress tolerance, growth, and access to nutrients in regolith/regolith simulant. • Potential pathways of human pathogenesis from plant microbiomes and how to test those pathways.
	For Year 1, seven projects have been selected across the three thrusts of the consortium. Efforts are collaborative across all three partners, with work planned to take place over the next few months and completing at the start of the second year of the consortium, based on academic calendar timelines. Thrust 1: Developing Biological Monitoring Capabilities Advanced health monitoring technologies based on flexible biosensor arrays. NASA’s upcoming missions beyond low Earth orbit (e.g., Artemis) present significant health risks due to prolonged exposure to ionizing radiation, microgravity, and circadian disruption. These stressors can lead to DNA damage, immune dysfunction, and metabolic imbalances, all of which threaten astronaut health and mission success. This seed project combines two projects from the initial proposal (1.1 and 1.2). This integrated project will focus on the development of miniaturized, wearable biosensors for real-time physiological health monitoring and radiation exposure assessments. The work initiated this year is focusing on metabolic/circadian biomarkers with ionizing radiation exposures, to be conducted at the Pacific Northwest National Laboratory (PNNL) in FY26 to test low dose ionizing radiation (IR)-relevant biomarkers. Internal desynchrony as a marker of ill health due to circadian misalignment. Astronauts often experience misalignment between endogenous circadian cycles and externally imposed behavioral schedules, leading to internal desynchrony, metabolic disturbances, and commensurate risks to their health and mission success during spaceflight. The gut microbiome plays significant roles in metabolism, may mediate internal desynchrony, and may offer avenues for developing effective medical countermeasures. As a step toward this technology, this seed project will characterize endogenous circadian rhythms of the gut microbiome and host peripheral oscillators in rodents under experimental conditions of circadian misalignment. The study aims to identify plasma biomarkers of internal desynchrony arising from misaligned feeding schedules, using metabolomics and lipidomics data measured by mass spectrometry (LC-MS/MS) in biobanked mouse liver samples from the participants recent study. Additionally, the project will investigate effects of misaligned feeding schedules on the activity of gut microbial enzymes (e.g., β-glucuronidase, cellulase, protease, and lipase) in biobanked gut content. As material for this project is already biobanked, the proteomics analyses will take place at PNNL in FY26. Characterizing plant cellular self-protection pathways in response to space-relevant abiotic stress. Space radiation exposures represent a significant challenge for plant development, as it increases production of reactive oxygen species

(ROS), causing oxidative damage to key biological molecules, compromising photosynthesis, and inducing premature senescence. ROS production also reduces synthesis of photo-assimilates, impacting essential plant-microbial interactions. Plant growth also depends on other environmental characteristics – including temperature, light, and soil – that have already been reproduced in the International Space Station (ISS). However, plant growth chambers are prone to malfunction, leading to heat stress and reduction of soil moisture content causing the drought stress. The longer-term goal of this project will develop remote sensors for measuring peroxisomes and autophagy on a cellular scale and use them to determine how combined stresses affect these cellular processes. The project will characterize cellular self-protection pathways in response to stress and determine the contribution of autophagy and peroxisomes to plant stress tolerance. The project will also determine how autophagy and peroxisomes contribute to interactions between plant beneficial and pathogenic microbes under stress and identify cellular responses to stress in Arabidopsis thaliana mutants deficient in autophagy, peroxisome biogenesis, and ROS scavenging. Initiated experiments are focusing on characterizing self-protection mechanisms in response to multiple stresses and will integrate ionizing radiation exposures at PNNL in FY26.

Thrust 2: Building System Models to Predict Robustness and Interactions in Space

Exploring respiratory epithelium homeostasis and response to stressors during space travel. Under spaceflight conditions, circulating levels of the cytokine hormone leptin increases (in part, due to physical inactivity and simulated weightlessness), and leptin expression in many tissues have been shown to be upregulated in hypoxic conditions in many models. While actions of this hormone are primarily associated with energy balance regulation, leptin signaling is known to upregulate mucus secretion in human bronchial epithelial cells in vitro, which could exacerbate respiratory health via mucus hypersecretion during spaceflight. However, leptin's role in modulating mucus secretion in the nasal, tracheal, or lung mucociliary epithelial linings in response to space-related stressors has not been studied. This project is investigating the role of leptin signaling in mucus secretion response to two main space-related stressors (hypoxia and ionizing radiation conditions) to test the hypothesis that upregulation of leptin signaling in response to these stressors is associated with mucus hypersecretion of the respiratory tract. The project will use the embryonic epidermis of the frog Xenopus tropicalis as an in vivo model of respiratory mucociliary epithelia (MCE), as it has been established as a powerful experimental model to study leptin regulation of MCE of the mammalian respiratory tract. The Xenopus embryonic MCE model shares cell types with human respiratory MCE, allowing in vivo study of heterogeneous cell interactions pertaining to MCE function and their interaction with resident microbiota, and is highly amenable to environmental and genetic manipulation experiments for proof-of-concept research. The project will: 1) Optimize methods to assess microbial community assembly on the surface of the Xenopus MCE. 2) Characterize leptin and leptin receptor expression profiles, mucus secretion, reactive oxygen species (ROS) response, and microbial communities in response to hypoxia and ionizing radiation. 3) Investigate modulation of leptin signaling and outcomes by exogenous leptin additions to culture media, CRISPR/CAS9 leptin knockdown embryos, and transient knockdown by immunoneutralization of leptin, through assessment of effects on mucus secretion, microbial community assembly, and performance trade-offs (e.g., oxygen consumption, growth and development rates). Work underway is focusing on the first subobjective and will integrate ionizing radiation exposures at PNNL in FY26. A Washington state invasive species permit has been secured to allow the Washington State University (WSU) team to transport Xenopus embryos to PNNL for low dose X-ray irradiations that expires on December 31, 2025 (the WSU and PNNL teams will work to renew this permit for CY26 transport and exposures).

Task Progress:

Effect of low dose ionizing radiation on kidney proximal tubule epithelium. Studies conducted aboard the ISS have shown that the space environment presents several risks and stressors, including microgravity and ionizing radiation (IR), both of which may affect kidney function. A better understanding of the effects of low dose charged particle and photon irradiation (protons and X-rays) on kidney proximal tubule structure and function will provide insight into the mechanisms of kidney dysregulation and guidance on mitigating risks of kidney injury and kidney stone formation. This project is using a flow-mediated microphysiological system (MPS), cultured with primary human proximal tubule epithelial cells irradiated with low doses (0.5 and 1 Gy) of 50 MeV protons, delivered at the University of Washington (UW) medical cyclotron facility, to simulate the space radiation environment. Matched X-ray exposures will also be conducted to establish relative biological effectiveness (RBE) values of the proton irradiations, with exposures conducted at both UW and PNNL in FY26. Cell viability will be measured using live/dead assays and biomarkers of injury and inflammation [kidney injury molecule-1 (KIM-1), interleukin-6 (IL-6)] determined at 12 h, 24 h, and 48 h post-irradiation timepoints. Transporter proteins (OAT1/3) will be visualized using immunocytochemistry to determine polarity.

Impacts of space and regolith stresses on the plant microbiome and proteome. Based on the fossil record, it is known that, through the symbiosis of plants with beneficial microorganisms, that plants were able to establish effectively on land. Plant-associated microorganisms can break down rocks into soil, increase the bioavailability of macronutrients including phosphate and potassium, fix atmospheric nitrogen gas to usable forms, reduce phytotoxicity of metals and organic pollutants, increase host plant tolerances to drought and temperature extremes, increase photosynthetic efficiency, and overall increase plant health under stressful conditions. Early successional plant species depend upon microbial symbioses for establishment under inhospitable conditions, serving as a useful source for impactful bacterial endophytes living within plants. Primary ecological succession can be thought of as the conversion of abiotic elements from the geosphere, hydrosphere, and atmosphere to the biosphere. To support any colony on Mars or the Moon, elements must either be converted from abiotic to biotic in situ or be brought from Earth. On Earth, endophytes are crucial for this conversion. They can convert poorly bioavailable phosphate and potassium insoluble minerals into more bioavailable organic and soluble potassium and phosphate. We will establish a collaborative study on the plant microbiome using their plant samples from spaceflight and ground controls. In addition to the microgravity and radiation stress of spaceflight, plants ultimately grown on lunar and Mars bases will require tolerance to the phytotoxicities of the corresponding regolith. Growth in these substrates will likely induce profound changes at the molecular level as the plant attempts to adjust. Studies have identified transcriptional changes associated with salt, metal, and ROS toxicity, as well as phosphate starvation in plant transcriptomics experiments in real lunar regolith. The project will grow plants inoculated with our known consortiums in regolith simulants and prepare samples for proteome; and, if funding permits, metabolome analysis to investigate how the host plant and microbiome respond. The project will examine what metabolites plants excrete in response to toxic metals such as chromium and aluminum. Initial experiments are growing plants with and without microbes and collecting samples for future proteomic analyses at PNNL in FY26.

Thrust 3: Apply Understanding of Human-Plant-Microbe Interactions to Enhance Health in Space

Developing molecular control systems for space-relevant plant chassis. NASA faces significant challenges in achieving self-sufficiency during long-term space missions. Key issues include production of on-demand medicines, monomers for 3D printing, and production of nutrient dense foods with potential to respond to individual astronauts' emerging needs, all under stressful spaceflight conditions with high CO2 and ionizing radiation. This project will build on our experiences with duckweed production and apply omics methods to non-models with and without relevant microbiomes under space emulating conditions, especially high CO2, which can reduce plant growth rates and nutritional value. In this initial funding period, the team will work together to advance our understanding of the responses of potentially space-relevant duckweed species through the following four tasks:

1) Conduct killing curve analysis of three diverse species of duckweed (e.g., Spirodella polyrhiza, Lemna minor, and Wolffia microscopia) towards identifying the more tolerant species to move forward with. Initial experiments will be conducted in sterile plates. Both vegetatively growing duckweed and dormant turions will be examined.

2) Conduct RNA-Seq under ionizing radiation of the top performing species from Task 1. RNA will be extracted from duckweed in a time course of exposure to ionizing radiation to understand how duckweed responds to this stress relative to other stress categories.

3) Improve duckweed genomic annotation and tools. The genome annotation of duckweed is superficial. We will focus on improving our understanding of the protein kinase complement of duckweed. Toward developing an understanding of gene regulation in duckweeds through the massively parallel reporter assay, plant STARR-Seq, we will begin testing protocols for duckweed protoplasting and transformation.

4) Implement kinase activity-based protein profiling in duckweed under ionizing radiation in parallel with Task 2. We will seek to identify the active kinases that are critical for ionizing radiation responses in duckweed, using the kinase activity-based protein probe, XO44. Initial experiments in plants by the Bartley and Lin Labs are underway now and we expect this important technology to be ready to apply to duckweed in the project period. Pilot experiments involve labeling of whole plants and extracts with XO44, followed by click labeling of the probe for visualization on a fluorescent gel. Then, proteomics will be conducted to identify active kinases in the same samples that are being characterized in Task 2 via RNA Seq. These kinases are key targets for design of protection strategies in future project periods. Tasks underway are focused on establishing duckweed growth at the WSU partner lab and testing the ABP labeling with gels for future proteomic analyses at PNNL.

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